

Take-Home Assignment

Predicting Private-Company Transactions from Public Data

1. Context

Third Bridge produces primary research - expert interviews, company briefs, and more - for private equity, credit, and corporate clients. Our clients care most about a company when it is about to be involved in a transaction: a buyout, a sale, an IPO, a refinancing. The earlier we know which companies those are, the earlier we can build the right content and be ready when the demand arrives.

Today, much of that prioritisation is reactive. We would like to make it predictive - and at scale. Imagine a function that continuously scans a large universe of private companies and surfaces the ones most likely to enter a transaction in the year ahead, so we can prioritise our coverage accordingly.

2. The challenge

We would like you to design - and, if you wish, partially prototype - an AI-powered tool that:

- Covers a universe of roughly 100,000 private companies (for the scope of this exercise, focus on 100 companies, but be prepared to answer how you could scale this to 100k companies)
- Uses only publicly available data, and
- Identifies the companies most likely to enter a transaction within the next ~12 months.

In short: from public signals alone, which private companies are heading towards a deal next year, and how confident are you?

3. What we'd like you to do

This is a design-led assignment. We do not expect a working system. We want to see how you would approach the problem, the judgement behind your choices, and how you would make it real and useful at scale. You should be prepared to defend your assumptions.

We would expect you to have a clear view on:

- **Definition.** What counts as a “transaction” for this purpose, and why.
- **Signals.** Which publicly observable signals would indicate a company is heading towards one.
- **Data.** Where you would source those signals, and how you would handle coverage and quality across 100,000 companies.
- **The role of AI.** Where language models or other AI methods add the most value - for example in extracting signals from unstructured text, enriching company profiles, or scoring - and where simpler methods are the better choice.
- **Scale and cost.** How you would make this run affordably over a large universe, rather than just a handful of companies.
- **Evaluation.** How you would know whether your predictions are any good.
- **Risks.** The main limitations, failure modes, and things you would want to validate.

You decide how to weight these. We are looking for sharp judgement and clear reasoning, not coverage of every point.

4. Your deliverable

A maximum of three slides covering:

- **Your approach.** How you would build the function — the architecture, the signals, the role of AI, and how you would make it work at scale.
- **Your reasoning and learnings.** What you believe the strongest signals are and why, what you would expect to find, the hardest parts of the problem, and what you would need to validate. If you build a proof of concept, summarise what it showed here.
- **A product view (optional).** Basic prototype to show you are comfortable with prompting, nothing fancy.

Three slides is a hard cap; deciding what to leave out is part of the exercise. If you build a proof of concept, you may attach a short notebook or prompt sample as supporting material - this sits outside the three-slide limit and is optional.

5. A note on data

Use only publicly and legitimately available data and be mindful of sources' terms of use - we care about how our data is sourced. You do not need to acquire or process any data; describing your intended sources and how you would use them is enough.

6. How we'll assess it

We will be looking at:

- **Problem framing and judgement.** Did you turn an open-ended, ambiguous problem into a sensible, defensible plan?
- **Technical depth.** Do your proposed methods - AI and otherwise - make sense, and would they actually work at scale and at reasonable cost?
- **Practicality.** Have you thought about coverage, data quality, evaluation, and the unglamorous realities of making this real?
- **Written communication.** This carries as much weight as the substance. We want to see a complex, technical idea explained clearly and concisely to a senior, non-technical reader. Your slides should stand on their own, without you there to present them.
- **Product thinking.** Can you connect the build to something commercially useful, with a clear sense of how success is measured.

7. A few practical notes

- **Effort.** This is designed to take few hours of focused work. We would rather see strong judgement on a well-chosen approach than exhaustive treatment.
- **Tools.** You are welcome to use Claude or similar tools to help structure and polish your slides. What matters to us is that the thinking, the conclusions, and the recommendation are genuinely your own - the tool is for production, not for judgement.
- **Imperfection is expected.** Assumptions, open questions, and “here is what I would do next” are all welcome. Tell us what you do not yet know.